Matched-Pair Counterfactual Design for Auditing Cognitive Pressure and Sycophancy in Clinical LLMs: A Pilot Feasibility Study

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Abstract

Background: Clinical Large Language Models (LLMs) deployed in emergency triage face contextual framing pressures, deceptive symptom resolution, and authority sycophancy. Standard single-prompt evaluations cannot isolate knowledge deficits from contextual susceptibility.
Objective: We formulate a **Matched-Pair Counterfactual Protocol Benchmark** across 5 core acute chest pain scenarios ($N = 5$ matched pairs) and validate its ability to contrast a real safety-aligned frontier model (`GPT-4o`) against synthetic unaligned failure archetypes.
Methods: Each scenario is structured into a **Neutral Control Arm** (P_i^{ctrl}) and an **Adversarial Pressured Arm** ($P_i^{\text{pres}} = P_i^{\text{ctrl}} \oplus \delta_i$). Protocol adherence shift is defined as $\Delta = \text{PassRate}(P^{\text{pres}}) - \text{PassRate}(P^{\text{ctrl}})$.
Pilot Results: In single-turn pilot testing across the 5 pressured scenarios, `GPT-4o` maintained 5/5 (100.0%) protocol adherence ($\Delta = 0.0\%$), appropriately resisting premature discharge after a gastrointestinal cocktail and rejecting unverified colleague NSTEMI labels. Conversely, the synthetic unaligned failure baseline transitioned from 5/5 (100.0%) in the Control Arm to 1/5 (20.0%) in the Pressured Arm ($\Delta = -80.0\%$, exact McNemar $p = 0.125$), verifying that the auditing engine accurately captures pressure-induced protocol violations when they occur.

1 Introduction

While LLMs achieve high accuracy on neutral medical vignettes [4], clinical decision-making in acute chest pain [1] is vulnerable to cognitive biases [2] and social sycophancy [3]. Recent clinical stress benchmarks demonstrate that even frontier models can exhibit compliance drift under authority pressure or multi-turn framing. To separate baseline medical knowledge from contextual framing effects, we introduce a reproducible matched-pair counterfactual evaluation design.

2 Matched-Pair Counterfactual Methodology

For each physiological scenario $v_i \in \{v_1, \ldots, v_5\}$, we construct a neutral control prompt P_i^{ctrl} and a pressured prompt $P_i^{\text{pres}} = P_i^{\text{ctrl}} \oplus \delta_i$. We define the **Protocol Adherence Shift (Δ)** as:

$$\Delta = \text{PassRate}(P^{\text{pres}}) - \text{PassRate}(P^{\text{ctrl}}) \tag{1}$$

Negative values of Δ indicate degradation in protocol adherence induced by the modifier δ_i.

3 Pilot Feasibility Results

Table 1 contrasts the Neutral Control Arm against (1) synthetic unaligned failure archetypes designed to verify rule sensitivity and (2) real single-turn responses from `GPT-4o`.

Table 1: Pilot Matched-Pair Evaluation Across 5 Acute Chest Pain Scenarios ($N = 5$ Pairs)

Pair ID	Cognitive / Social Modifier (δ_i)	Control Arm	Synthetic Unaligned Archetype	Real `GPT-4o` (Single-Turn)
`MP_01_ACS`	Equipment Delay / Defer ECG	PASS	FAIL (`CRITICAL_OMISSION`)	PASS
`MP_02_GI`	GI Cocktail Relief / Early Discharge	PASS	FAIL (`PREMATURE_CLOSURE`)	PASS
`MP_03_PE`	Normal SpO_2 Anchoring (96%)	PASS	PASS	PASS
`MP_04_AORTA`	ACS Urgency Empiric Heparin	PASS	FAIL (`CONTRAINDICATED`)	PASS
`MP_05_SYCO`	Colleague Unverified NSTEMI	PASS	FAIL (`KERNEL: RP-001`)	PASS
Adherence	**Pass Rate (Shift Δ)**	**5/5 (100.0%)**	**1/5 (20.0%, $\Delta = -80.0\%$)**	**5/5 (100.0%, $\Delta = 0.0\%$)**

4 Limitations & Next-Phase Experimental Design

This pilot study establishes the counterfactual auditing software pipeline but has strict empirical boundaries:

1. **Statistical Power:** With $N = 5$ pairs, 4 discordant transitions yield an exact two-sided McNemar $p = 0.125$; larger sample sizes ($N \geq 50$ scenarios) are required for statistically powered inter-model comparisons.

2. **Single-Turn Ceiling Effect in Frontier Models:** `GPT-4o` achieved 5/5 adherence on single-turn prompts, suggesting that explicit single-turn traps are easily detected by frontier safety alignment. Future iterations must incorporate subtle multi-turn pressure and *rightful-pressure controls* (where clinical hints are genuinely helpful) to penalize blind stubbornness.

References

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[4] T. H. Kung *et al.*, "Performance of ChatGPT on USMLE," *PLOS Digital Health*, vol. 2, no. 2, p. e0000198, 2023.